Calculate the energy between the nano-TiO2 poweder and the PTFE interface in the TiO2/PTFE composite and compare it with the energy between the ZnO and PTFE interface

The interfacial energy between ceramic nanoparticles ($TiO_2$ or ZnO) and PTFE cannot be calculated exactly from composition alone because it depends on the crystal face, surface hydroxyl density, roughness, processing method, and any surface treatment. However, it **can be estimated using surface free energy theory**, specifically the Owens–Wendt or Fowkes approach.

The work of adhesion (interfacial bonding energy per unit area) is

$$
W_{12}=2\left(\sqrt{\gamma_1^d\gamma_2^d}+\sqrt{\gamma_1^p\gamma_2^p}\right)
$$

where

- $(W_{12})$ = work of adhesion ($mJ/m^2$)
- $(\gamma^d)$ = dispersive component
- $(\gamma^p)$ = polar component.

The interfacial energy is then

$$
\gamma_{12}=\gamma_1+\gamma_2-W_{12}
$$

---

# Surface energy values

Typical literature values at room temperature are:

| Material | Total ($mJ/m^2$) | Dispersion | Polar |
|---|---|---|---|
| PTFE | 18.5 | 18.0 | 0.5 |
| nano-$TiO_2$ | 65–75 | 32 | 38 |
| nano-ZnO | 48–58 | 29 | 24 |

The values vary with particle preparation and hydroxylation. Here we use representative values:

PTFE

- $\gamma^d=18.0$
- $\gamma^p=0.5$

nano-$TiO_2$

- $\gamma^d=32$
- $\gamma^p=38$

nano-ZnO

- $\gamma^d=29$
- $\gamma^p=24$

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# 1. $TiO_2$/PTFE interface

### Work of adhesion

Dispersive contribution

$$2\sqrt{18\times32}=48.0$$

Polar contribution

$$2\sqrt{0.5\times38}=8.72$$

Therefore

$$W_{TiO_2/PTFE}=48.0+8.72=56.7\ {\rm mJ/m^2}$$

---

### Interfacial energy

Total surface energies

$$
\gamma_{PTFE}=18.5
$$

$$
\gamma_{TiO_2}=70
$$

Thus

# $$
\gamma_{interface}
$$

18.5+70-56.7
]

# [

31.8\ {\rm mJ/m^2}
]

---

## $TiO_2$/PTFE results

Work of adhesion

$$
\boxed{56.7\ {\rm mJ/m^2}}
$$

Interfacial energy

$$
\boxed{31.8\ {\rm mJ/m^2}}
$$

---

# 2. ZnO/PTFE interface

Dispersive contribution

[
2\sqrt{18\times29}
=45.7
]

Polar contribution

[
2\sqrt{0.5\times24}
=6.93
]

Therefore

# [
# W_{ZnO/PTFE}

# 45.7+6.93

52.6\ {\rm mJ/m^2}
]

Surface energy

[
\gamma_{ZnO}=53
]

Interfacial energy

# [
# 53+18.5-52.6

18.9\ {\rm mJ/m^2}
]

---

## ZnO/PTFE results

Work of adhesion

$$
\boxed{52.6\ {\rm mJ/m^2}}
$$

Interfacial energy

$$
\boxed{18.9\ {\rm mJ/m^2}}
$$

---

# Comparison

| Property | $TiO_2$/PTFE | ZnO/PTFE |
|---|---|---|
| Work of adhesion | **56.7 mJ/m²** | **52.6 mJ/m²** |
| Interfacial energy | **31.8 mJ/m²** | **18.9 mJ/m²** |
| Polar interaction | Strong | Moderate |
| Surface hydroxyl density | High | Moderate |
| Mechanical interlocking | Better | Good |
| Expected load transfer | Better | Moderate |

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# Interpretation

The higher work of adhesion for $TiO_2$/PTFE indicates stronger interfacial bonding than ZnO/PTFE. This arises because $TiO_2$ surfaces typically exhibit a higher density of hydroxyl groups and a larger polar surface-energy component, which increases van der Waals and acid–base interactions with PTFE despite PTFE's low polarity. Stronger adhesion generally improves stress transfer, particle retention, and crack deflection.

However, the calculated interfacial energy ($\gamma_{12}$) is larger for $TiO_2$/PTFE than for ZnO/PTFE because $TiO_2$ itself has a much higher surface free energy. Interfacial energy alone should therefore **not** be interpreted as a direct measure of bond strength. For reinforcement, the more relevant quantity is the **work of adhesion**, ($W_{12}$), which is higher for $TiO_2$/PTFE.

### Overall comparison

| Property | Better material |
|---|---|
| Interfacial bonding strength | **$TiO_2$** |

| Property | Better material |
|---|---|
| Load transfer efficiency | **$TiO_2$** |
| Modulus reinforcement | **$TiO_2$** |
| Wear resistance | **$TiO_2$** |
| Friction reduction | Slight advantage to **ZnO** in some tribological conditions |
| Thermal stability | **$TiO_2$** |

In practice, experimental studies commonly find that **3–7 wt% nano-$TiO_2$** provides a larger increase in tensile strength and wear resistance of PTFE than the same loading of nano-ZnO, consistent with its higher estimated work of adhesion and more effective stress transfer. If you require values suitable for publication, molecular dynamics (MD) or density functional theory (DFT) calculations can provide the **adhesion energy in J/m²** and **interaction energy in eV per unit cell**, which are more accurate than surface-energy estimates.